Fiscal Year:	FY 2014	**Task Last Updated:**	FY 10/27/2013
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2012	**End Date:**	02/28/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2017 per NSSC information (originally 9/30/2016)--Ed., 4/23/13		
Key Personnel Changes/Previous PI:	none.		
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX12AR32G		
Performance Goal No.:			
Performance Goal Text:			
	Electrically driven flows can provide significant enhancements to current and future technologies that can benefit from increased heat transfer and mass transport capacities in macro and micro scales. In particular, electrically generated flows (with and without phase change) have great potential for terrestrial and micro-gravity applications. Electrohydrodynamic (EHD) ion-drag pumping and induction pumping have been extensively studied in the past several decades with no significant, successful applications. On the contrary, EHD pumping based on conduction phenomenon has reached a stage where various applications are expected in the near future, since pressure heads on the order of kilo Pascal per electrode pair are easily obtainable. EHD conduction pumps are simple in design, lightweight, non-mechanical, and free of vibration and noise, and they allow for active control of heat transfer and mass transport. EHD conduction pumping requires minimal electric power to operate (at least three orders of magnitude less power		

Task Description:	compared to the resultant heat transport capacity). This proposed research will fundamentally investigate the electrically-driven dielectric liquid film in the presence of phase change in macro- and micro-scales based on conduction phenomenon through a combined experimental and computational investigation. Furthermore, the coupling of electric double layers and heterocharge layers in micro-channels will be explored, and the non-dimensional threshold at which the double layer starts to affect the corresponding flow field will be determined. Note that in micro-scales, the Debye length within dielectric fluids becomes comparable to the conduit dimensions. Additionally, unique two-phase heat transport devices will be developed that will operate based on the conduction phenomenon, with or without capillary effects, in macro- and micro-scales. Heat flux levels with these active micro-scale heat transport devices are expected to significantly exceed the corresponding working fluid critical heat flux level while operating stably and for example reaching much above 300 W/cm2 for refrigerant R-123.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are a number of potential terrestrial and space applications for this EHD driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure etc.). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices. For further technical information, please contact Professor Jamal Yagoobi at jyagoobi@wpi.edu . For general information, please contact Mr. Robert Anderson at anderson@iit.edu . This device currently is not commercially available.
Task Progress:	An innovative radial heat transport device was conceptualized, designed, fabricated, and tested (proof of concept) that used electrohydrodynamic (EHD) conduction pumping with concentric ring electrodes flushed with the bottom wall to pump a thin liquid film from a peripheral condenser to a small, centrally-located boiler. The performance of the device was experimentally characterized in the lab, and its behavior with EHD driven flow was compared to the case of an almost-stagnant liquid film where gravity was the only pumping force. EHD produced significant enhancements by ensuring a sufficient supply of bulk liquid to the boiler section. The significant circulating velocities produced by EHD cause a notable reduction in the superheat of the boiler surface for a given heat flux when compared to the no-EHD case where the liquid film is almost stagnant. This reduction in surface temperature can be highly beneficial in electronics cooling applications, since such electronic components are highly sensitive to operating temperature. A reduced operating temperature can improve the reliability of the electronic component, or the power density (and processing power) of the component can be increased, as long as the total heat flux generation is below the critical heat flux of the two-phase cooling process. In September 2013, Jamal Yagoobi (PI) and his PhD student (Viral Patel) along with Jeff Didion (Co-PI) and Frank Robinson of NASA GSFC tested the radial two-phase heat transport device on board of a reduced gravity flight. The results are currently being analyzed.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Articles in Other Journals or Periodicals	Patel VK, Robinson F, Seyed-Yagoobi J, Didion J. "Terrestrial and micro-gravity experimental study of micro-scale heat transport device driven by electrohydrodynamic conduction pumping." IEEE Transactions on Industry Applications. 2013; Early Access available online: http://dx.doi.org/10.1109/TIA.2013.2264042 (accepted for publication). , Oct-2013
Articles in Peer-reviewed Journals	Pearson MR, Seyed-Yagoobi J. "Electrohydrodynamic conduction driven single- and two-phase flow in microchannels with heat transfer." ASME Journal of Heat Transfer. 2013 Oct;135(10):101701. http://dx.doi.org/10.1115/1.4007617 , Oct-2013
Articles in Peer-reviewed Journals	Pearson MR, Seyed-Yagoobi J. "EHD conduction-driven enhancement of critical heat flux in pool boiling." IEEE Transactions on Industry Applications. 2013 Jul-Aug;49(4):1808-16. http://dx.doi.org/10.1109/TIA.2013.2262451 , Jul-2013
Papers from Meeting Proceedings	Patel V, Seyed-Yagoobi J. "Micro-Scale Two-Phase Flow Heat Transport Device Driven By Electrohydrodynamic Conduction Pumping." ASME 2013 Summer Heat Transfer Conference, Minneapolis, Minnesota, July 14-19, 2013. ASME 2013 Summer Heat Transfer Conference, Minneapolis, Minnesota, July 14-19, 2013. Paper HT2013-17196. , Jul-2013
Papers from Meeting Proceedings	Patel V, Seyed-Yagoobi J. "Long-term Performance Evaluation of Two-phase Heat Transport Device driven by EHD Conduction." IEEE-IAS Annual Meeting, Orlando, FL, October 6-11, 2013. IEEE-IAS Annual Meeting, Orlando, FL, October 6-11, 2013. Paper 2013-EPC-347. , Oct-2013